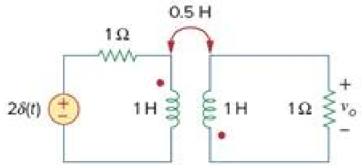
Chapter 18, Problem 50P

Problem

Determine *vo*(*t*) in the transformer circuit of Fig. 18.52.

Figure 18.52

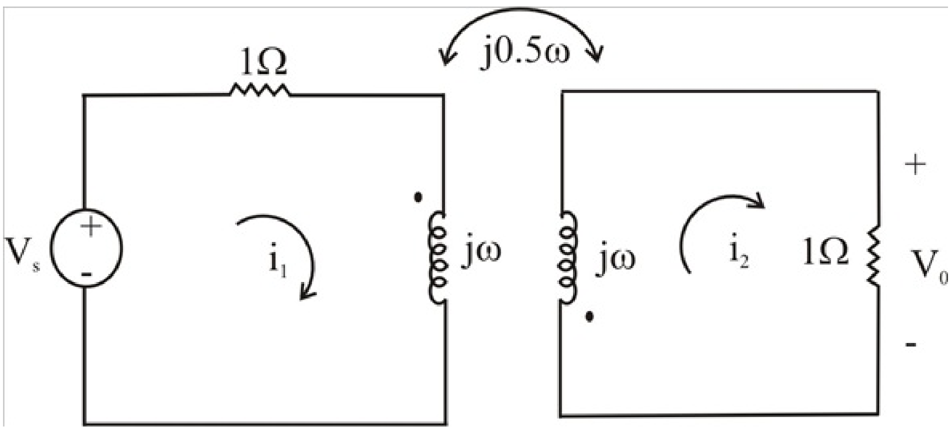


Step-by-step solution

Step 1 of 3

Consider the circuit shown below:-

Step 2 of 3



Step 3 of 3

$$\textit{for loop } 1. \quad -2+(1+j\omega)I_2+0.5j\omega I_2=0 \qquad -(1)$$

$$\textit{for loop } 2. \quad (1+j\omega)I_2+j0.5\omega I_2=0$$

$$I_1=\frac{(1+j\omega)I_2}{-j0.5\omega}=\frac{-2(1+j\omega)I_2}{j\omega}$$

Subsituting this into (1)

$$2=\frac{-2(1+j\omega)I_2}{j\omega}+\frac{j\omega I_2}{2}$$

$$2j\omega=-\left(4+j4\omega-\frac{3}{2}\omega^2\right)I_2$$

$$I_2=\frac{2j\omega}{\left(4+4j\omega-1.5\omega^2\right)}$$

$$V_o=I_2=\frac{-2j\omega}{4+j4\omega+\left(1.5j\omega^2\right)}$$

$$V_o=\frac{j\omega\frac{4}{3}}{\frac{8}{3}+j\frac{8\omega}{3}+(j\omega)^2}$$

$$=\frac{-4\left(\frac{4}{3}+j\omega\right)}{\left(\frac{4}{3}+j\omega\right)^2+\left(\frac{\sqrt{8}}{3}\right)^2}+\frac{\frac{16}{3}}{\left(\frac{4}{3}+j\omega\right)^2+\left(\frac{\sqrt{8}}{3}\right)^2}$$

$$v_o(t)=-4e^{\frac{4t}{3}}\cos\left(\frac{\sqrt{8}}{3}t\right)u(t)+5.657e^{\frac{-4t}{3}}\sin\left(\frac{\sqrt{8}}{3}t\right)u(t)\ \textit{Volts}$$